

Unbounded Wind Fields for Stratospheric Balloon Simulation

Lazy spatial queries from a bounded ERA5-conditioned diffusion model

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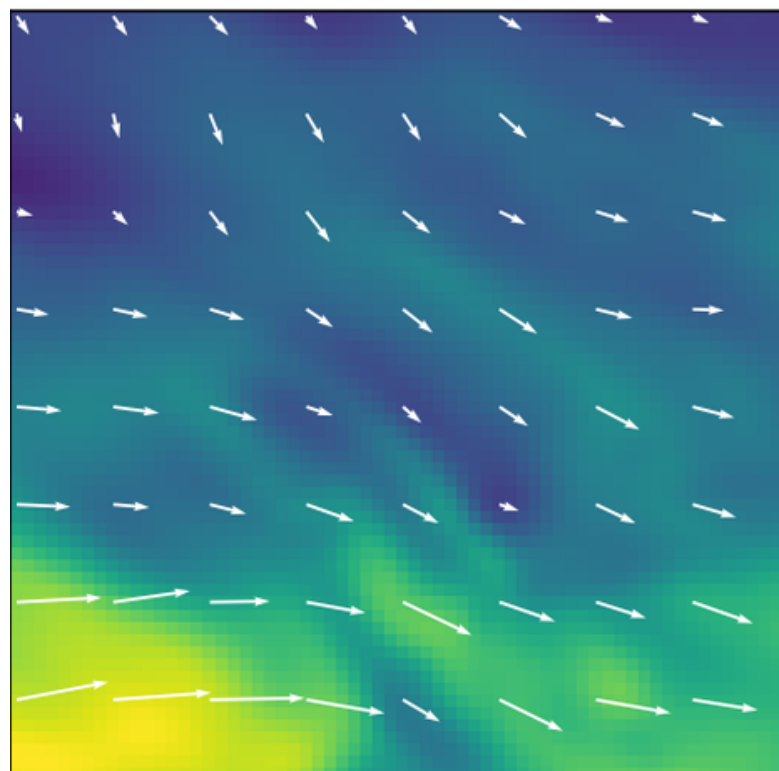
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github.com/sdean-group/balloon-env-dev

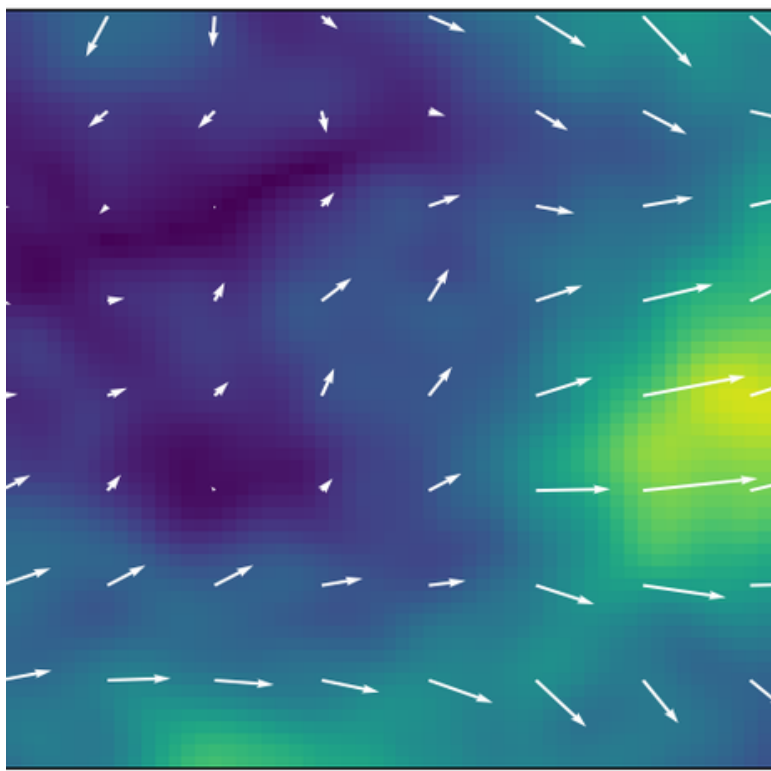
1 Why generate wind worlds?

Balloon simulators need winds that are both **realistic** and **effectively unbounded**.

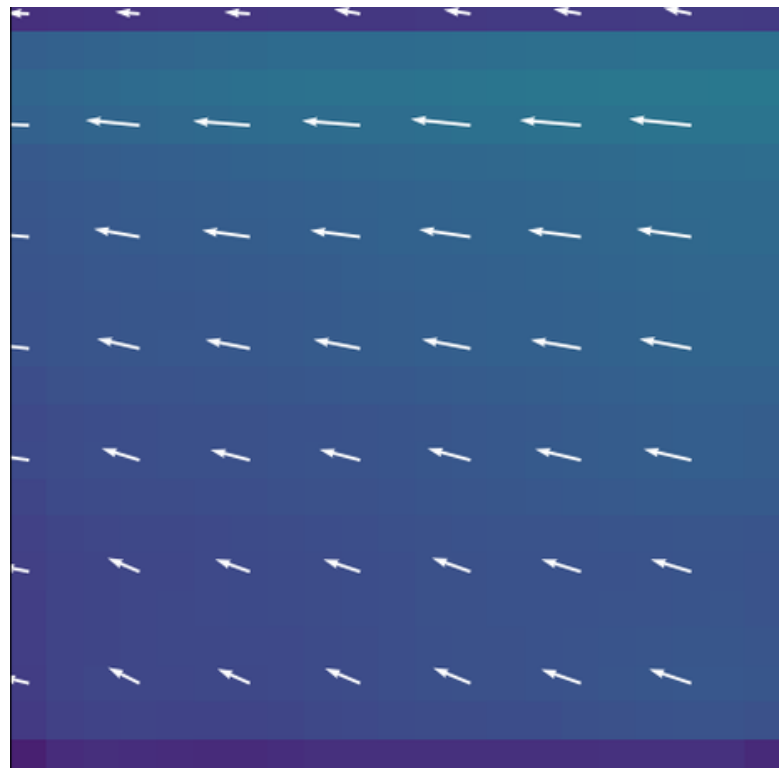
A balloon changes altitude to exploit winds with different speeds and directions. A finite weather archive encourages memorization; simple procedural fields can miss the structures that make altitude control useful.



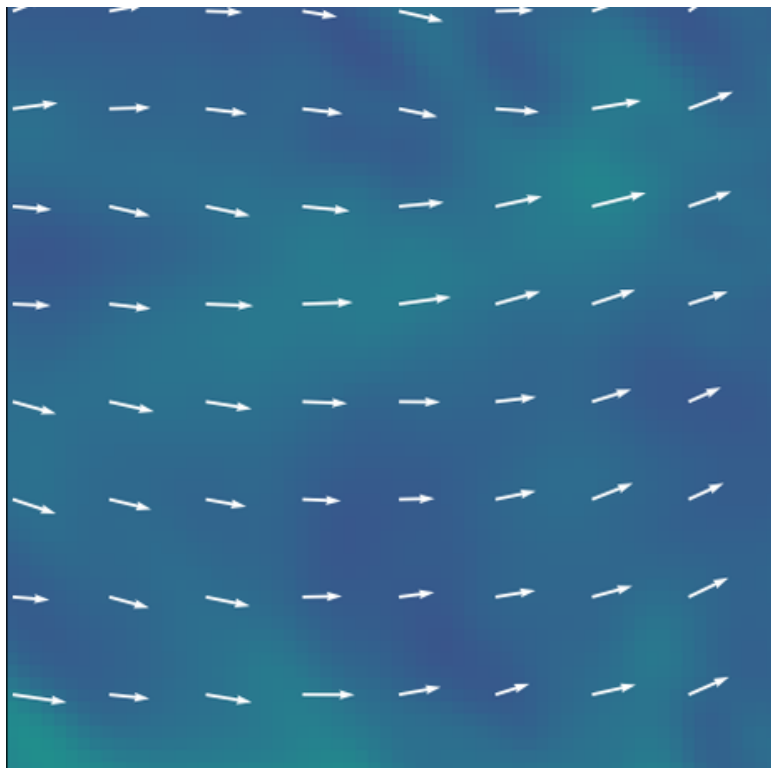
ERA5: realistic, finite archive



simplex: unlimited, fitted noise



BLE-VAE: learned, bounded

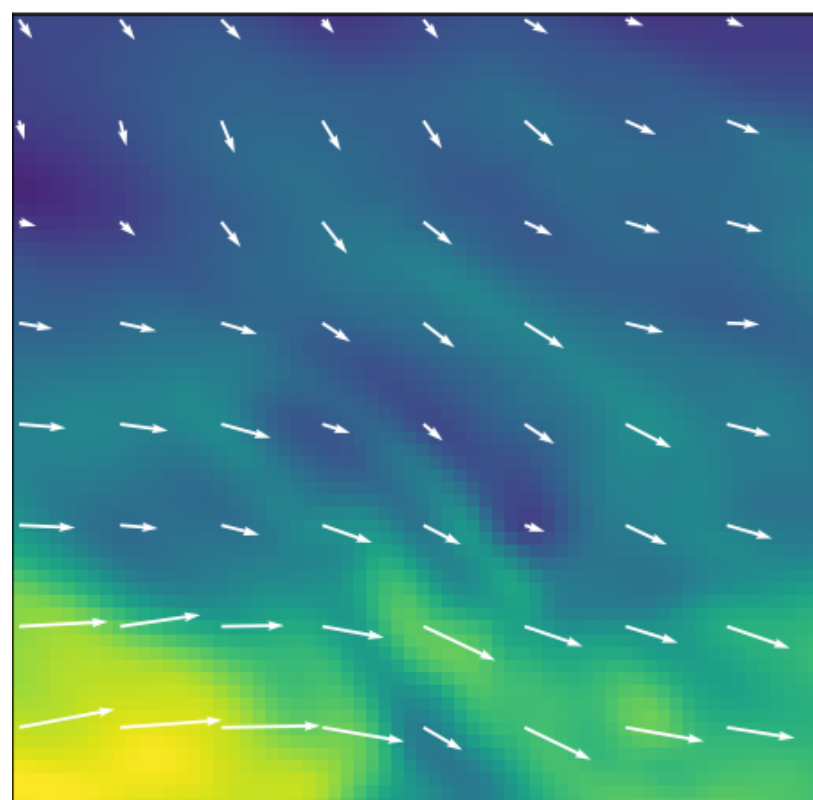


ours: learned, random access

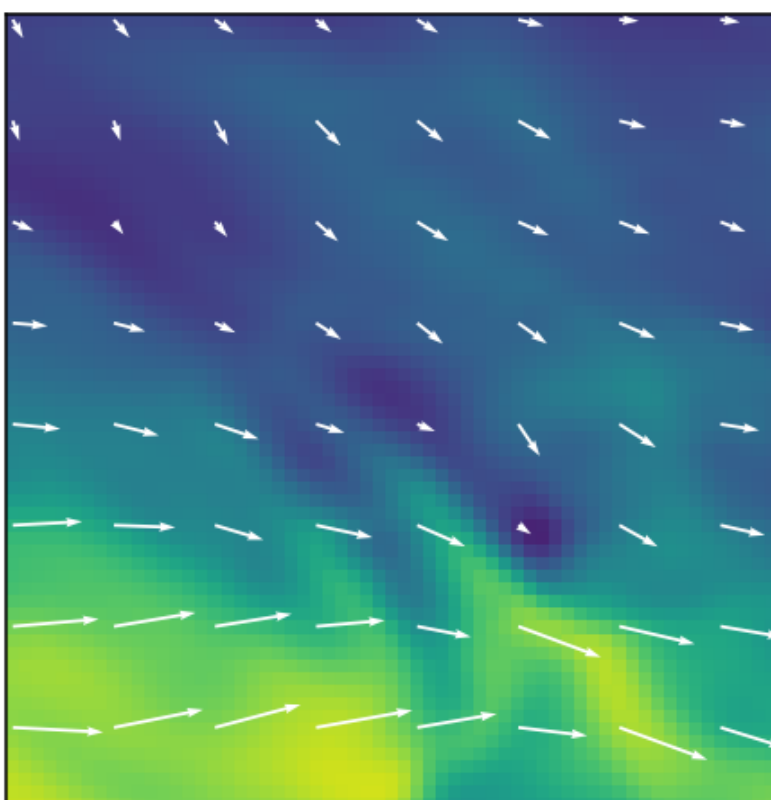
Our contributions

- A diffusion model for four hourly ERA5 wind volumes on 18 vertical levels.
- A lazy InfiniteDiffusion wrapper for reproducible, coordinate-indexed queries.
- A benchmark spanning spectra, marginals, extremes, vertical structure, and tiling.

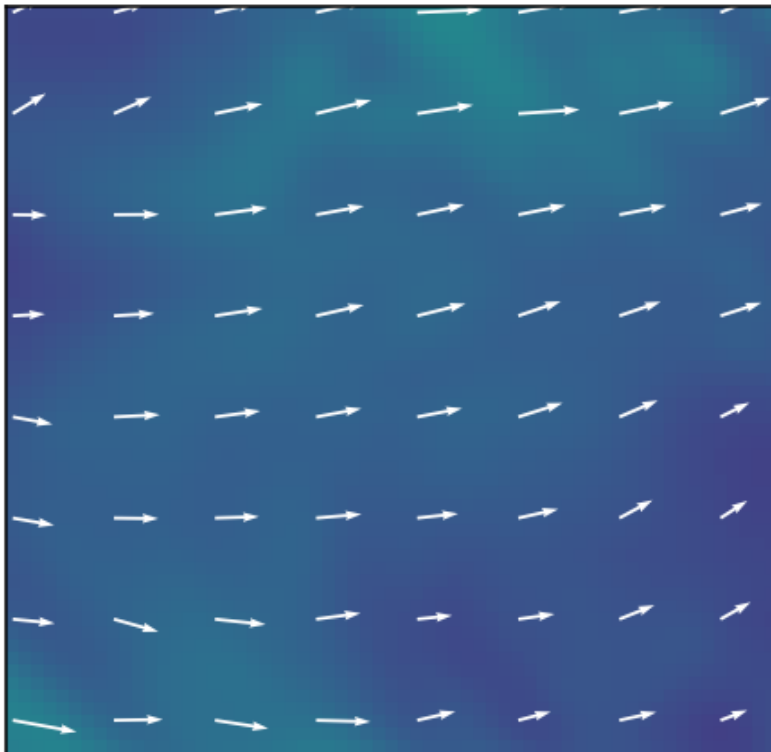
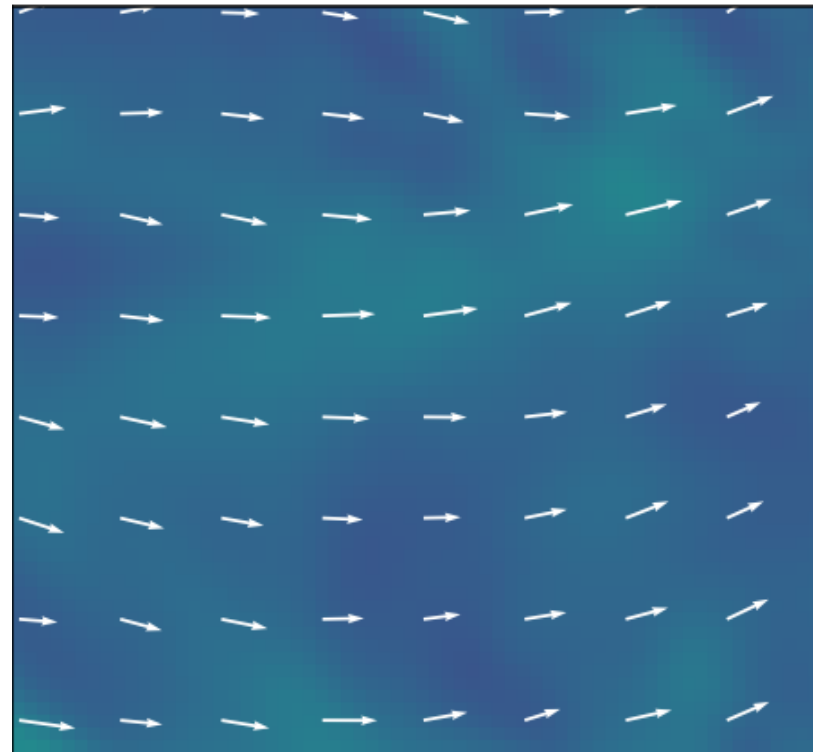
What does the base generator learn?



diff 300k+churn +0h max 19.5 m/s



diff 300k+churn +3h max 18.6 m/s

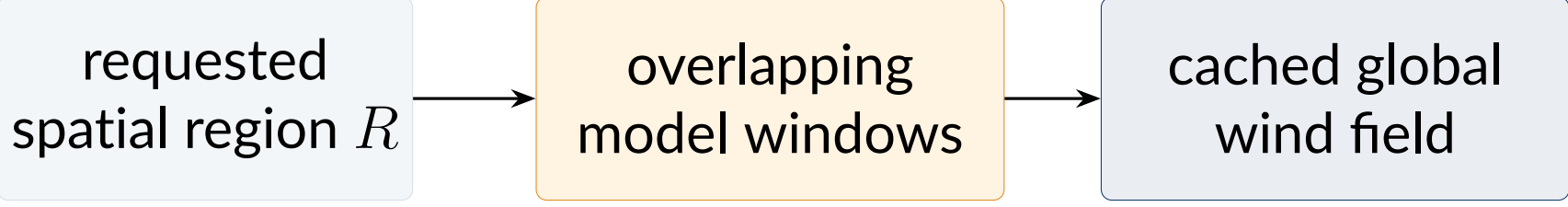


Same difficult January condition at ≈ 91 hPa. Top: ERA5; bottom: diffusion. Columns show +0 h and +3 h.

Visual read: the model produces coherent, evolving flow, but misses the strongest winter jet and loses fine-scale variation.

2 Bounded model, unbounded field

Train one local denoiser, then reuse it wherever a simulator asks for wind. Overlapping predictions are tapered, cached, and normalized globally.

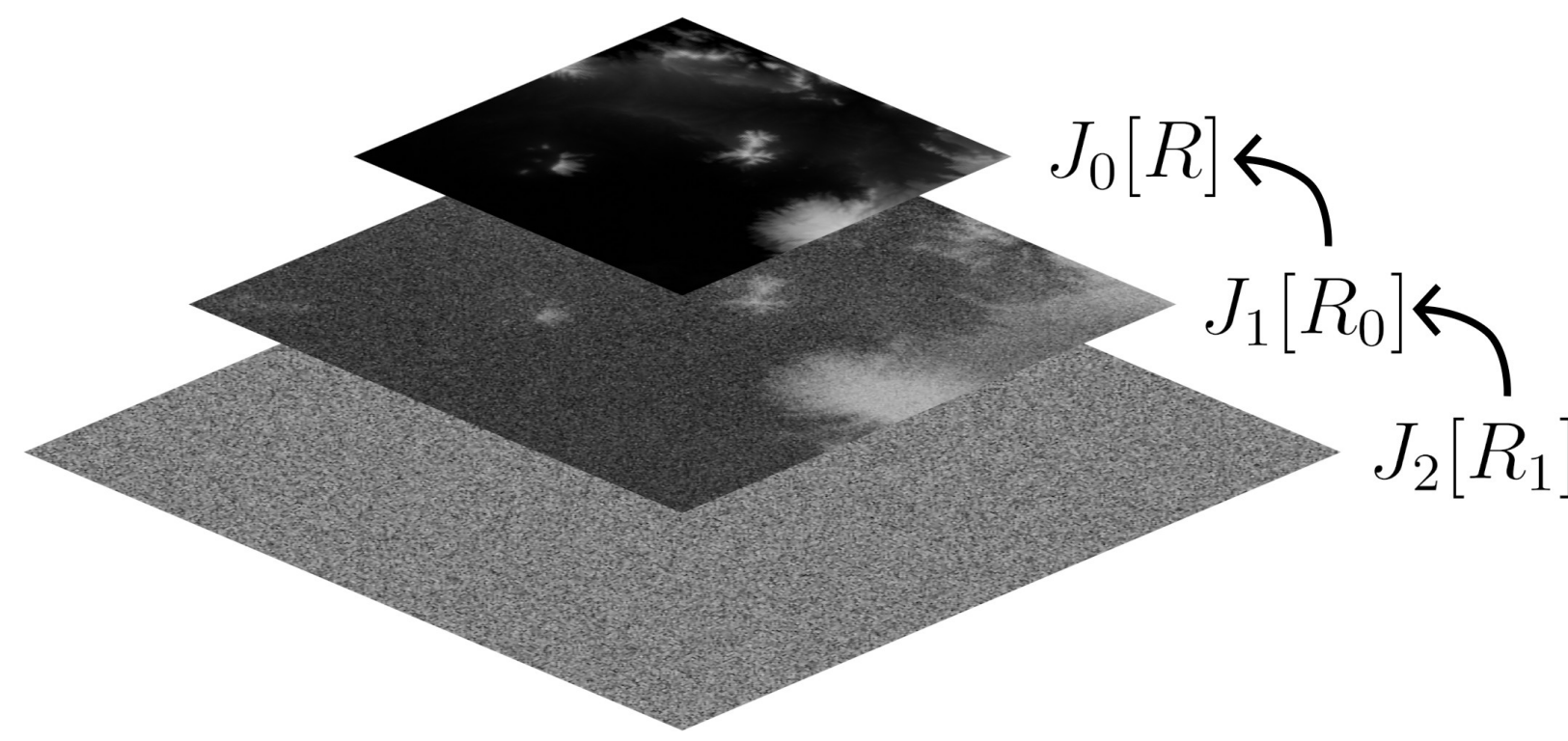


Adapted from Goslin et al., *InfiniteDiffusion*:

$$J_t[R] = \left[\frac{\sum_{i \in \kappa(R)} U_i(W_i \odot \Phi(J_{t+1}[R_i] | c_i))}{\sum_{i \in \kappa(R)} U_i(W_i)} \right] [R]$$

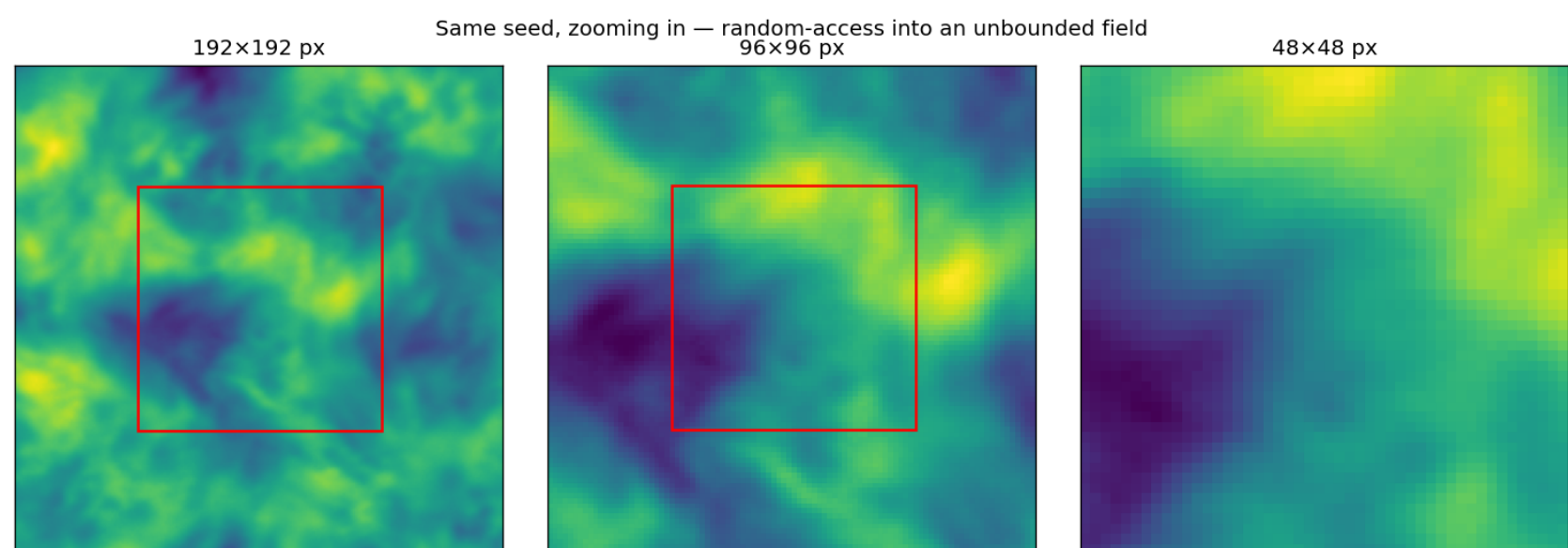
$\kappa(R)$ selects windows; U_i embeds local predictions; W_i downweights window edges.

Why a local query creates a dependency pyramid



Noise is evaluated over a larger parent region, then denoised into progressively smaller context and output regions. Reproduced from Goslin et al., Fig. 2.

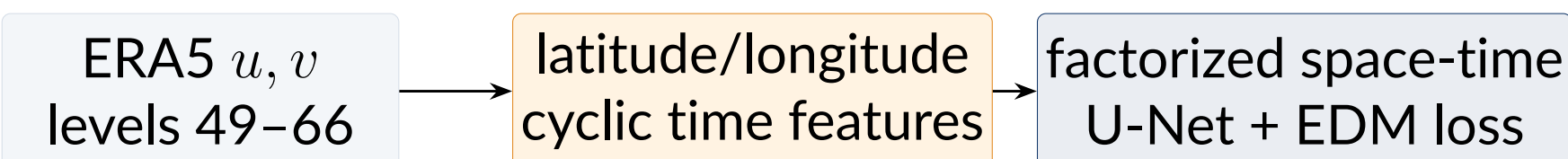
3 Reproducible random access



One seed defines one virtual wind world. Nested queries agree on shared coordinates and do not require generating the surrounding world first.

∞	$O(1)$	=
Unbounded	Random access	Seed-consistent
coordinates have no canvas edge	fixed-query cost ignores total world extent	same coordinates return the same field

Base denoiser and conditioning



Training objective: denoise at a randomly sampled noise level

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0, \sigma, \varepsilon} \left[\lambda(\sigma) \|D_\theta(x_0 + \sigma\varepsilon; \sigma, c) - x_0\|_2^2 \right]$$

- Four hourly 3-D wind volumes are denoised jointly.
- Location and year/half-year/day harmonics travel with every window.
- For a fixed-size spatial query, cost is independent of total virtual-world extent.

Scope of the completed system

Spatial queries are unbounded and seed-consistent. Time is generated in four-hour blocks; arbitrary-length temporal tiling is future work.

4 How we test realism

A realistic generator should reproduce held-out ERA5 in four distinct ways:

1. Distributions

Typical u , v and speed, plus rare jet-speed tails.

2. Spatial physics

Energy at each wavelength and the divergence/vorticity balance.

3. Vertical structure

Shear and winds that reverse direction across altitude.

4. Temporal physics

Persistence and transported-particle spread over long sequences.

Why this measures realism. Each metric is a distance to held-out ERA5; the ERA5 self-split floor shows natural sample-to-sample variation. Approaching that floor **across all four views** is stronger evidence than any single score, because fitted noise can imitate one statistic while producing physically wrong winds. Reference: days 8–14 of Jan/Apr/Jul/Oct 2023.

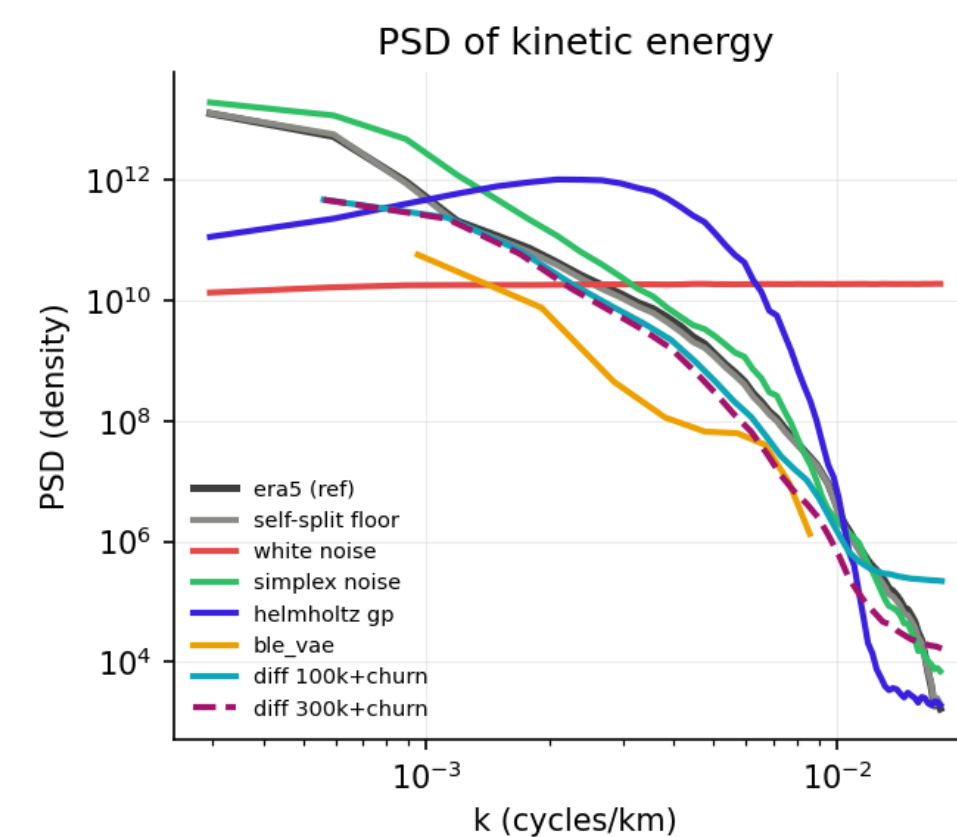
5 Results: extent solved, realism open

metric	floor	simplex	BLE-VAE	ours*
$SR_E \downarrow$	0.25	0.75	2.73	1.53
$SR_{div} \downarrow$	0.25	0.87	5.23	1.56
$SR_{vort} \downarrow$	0.25	0.72	2.16	1.53
$L_{eff} \text{ (km)} \downarrow$	56 [†]	56 [†]	842	337
$W_1(u \text{loc, time}) \downarrow$	2.08	4.69	N/A	3.00
$W_1(v \text{loc, time}) \downarrow$	2.67	2.08	N/A	4.04
tail error 0.1% $\downarrow$	2.92	2.34	18.94	13.54
jet elongation ratio $\rightarrow 1$	1.17	0.85	1.76	0.95

Main result

Condition-matched W_1 is computed across levels at fixed location, month, and hour. Ours beats simplex for u (3.00 vs 4.69), but not for v (4.04 vs 2.08): conditioning improves one component, not the full wind distribution.

*Table: four-year, 250k-step checkpoint (best overall checkpoint).



Earlier one-year sweep: diffusion follows ERA5 at large scales but loses fine-scale power. [†]Resolved across the compared range.

6 Takeaway and next steps

InfiniteDiffusion solves extent and reproducibility; the base denoiser remains the realism bottleneck.

Completed

- spatially unbounded queries
- deterministic revisits
- four-hour coherent blocks
- reference-based benchmark

Future work

- coarse synoptic conditioning
- stronger jet extremes
- arbitrary-length time tiling
- train and evaluate balloon agents